
Business Requirement Specification: Financial Services - Product Shop

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Glossary of Terms

Term	Definition
Product Eligibility	Criteria used to determine if the customer's profile meets the requirements to take up a given product
Fulfilment	The process of ensuring that the relevant checks have taken place in order to ensure that a product can be taken out by the person that applied for it

Glossary of Acronyms

Term	Definition
DHA	Department of Home Affairs
KYC	Know Your Customer

1 Project Overview

The project involves implementing a digital product shop to allow users to select and take-up products offered by a financial services institution. This encompasses profile creation, various eligibility checks, and fulfilment, and will require the integration of the solution into multiple existing systems.

2 Use Case Diagram

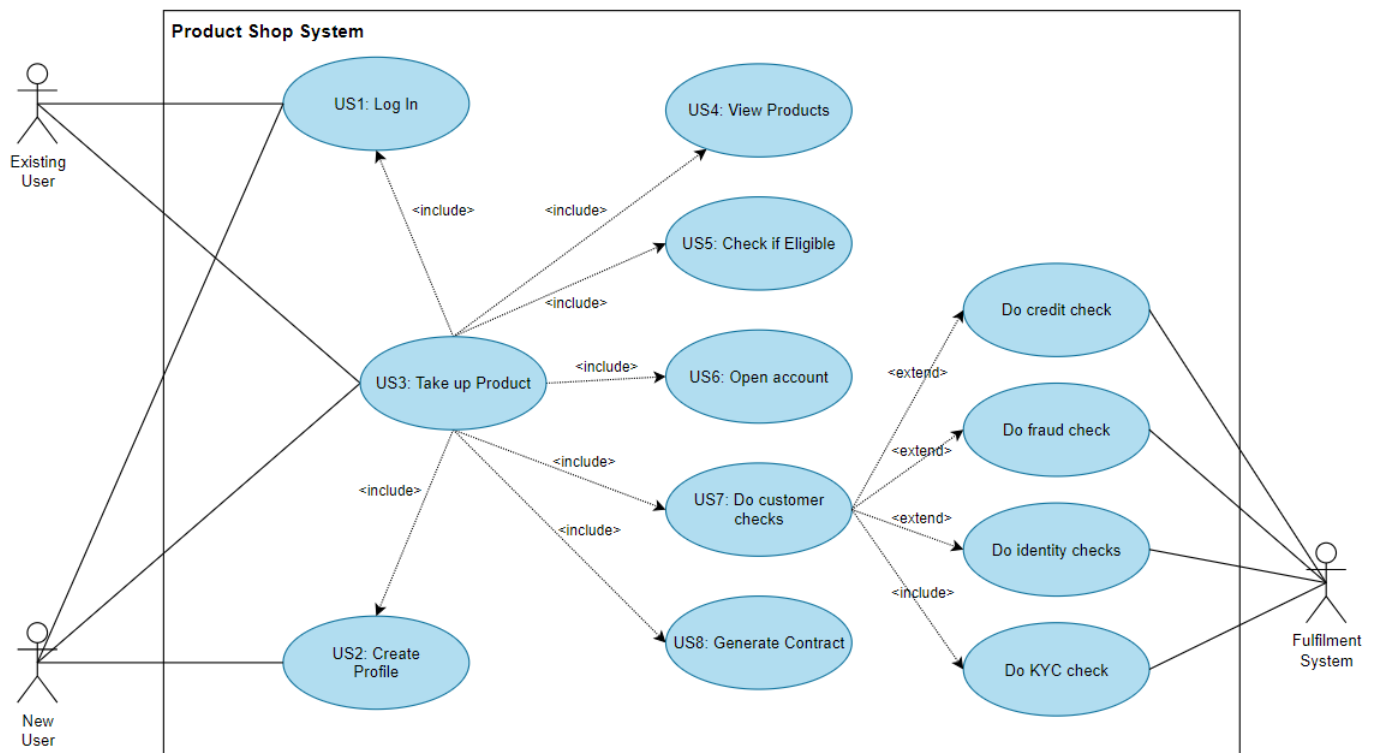
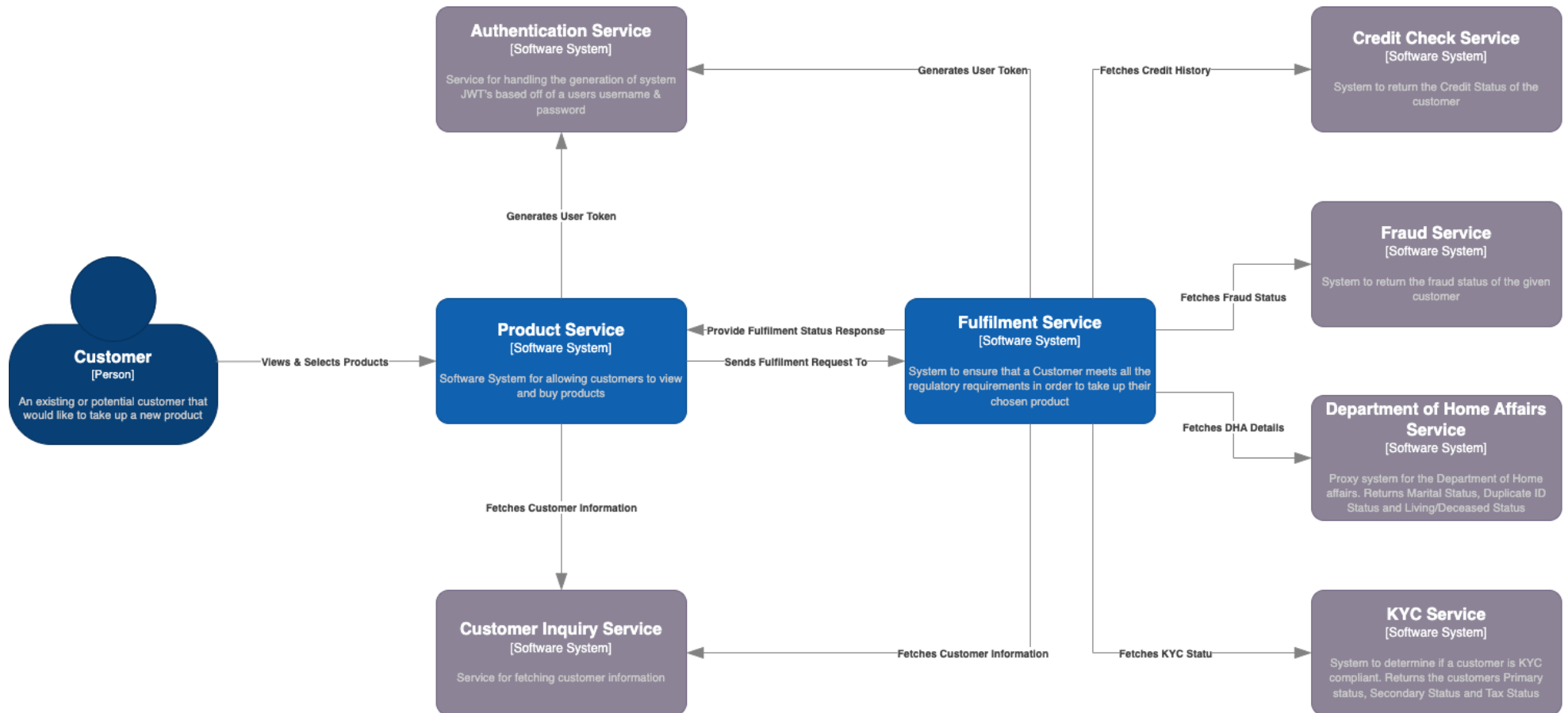


Figure 2-1 Use Case Diagram for the Product Sales Platform

3 System Diagrams

3.1 SYSTEM CONTEXT DIAGRAM



[System Context] Product Sales Platform
The system context diagram for the Product Sales Platform

Figure 3-1 System Context Diagram for the Product Sales Platform

3.2 SYSTEM CONTAINER DIAGRAM

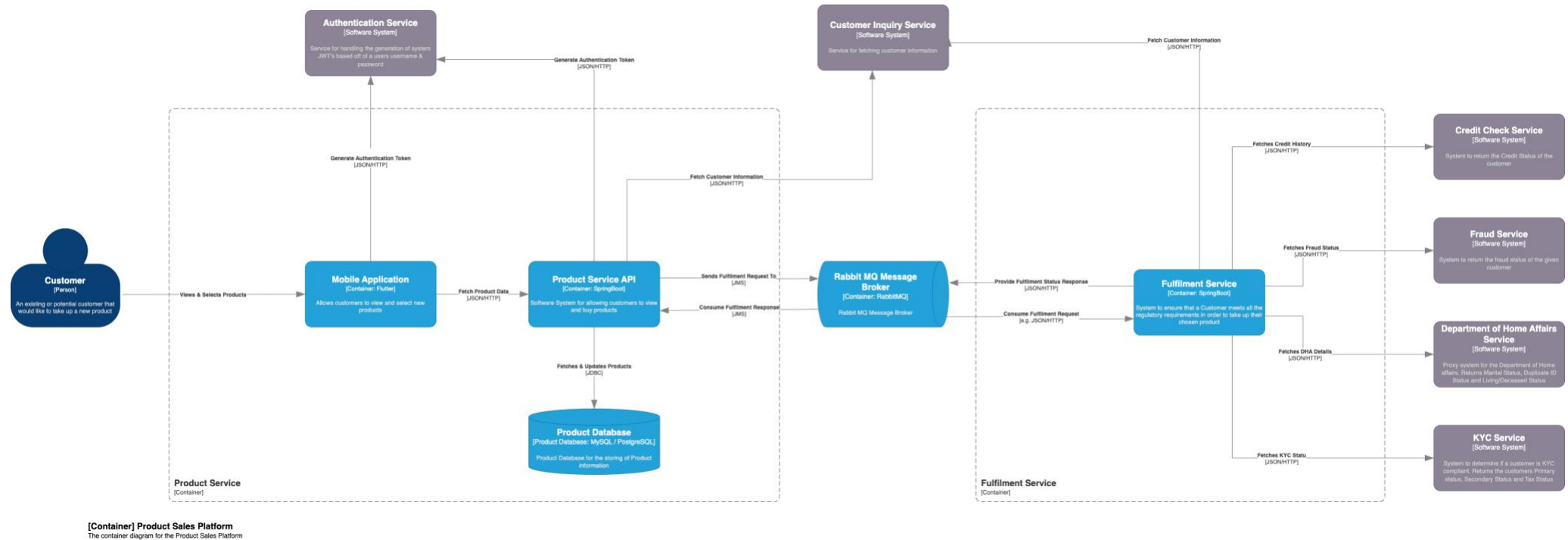


Figure 3-2 Container Diagram for the Product Sales Platform

4 User Stories

User Story	User Story Name	User Story Description
US1	Log In	As an existing user I want to log in so that I can browse and take up products
US2	Create Profile	As a new user I want to be create a profile so that I can browse and take up products
US3	Take up Product	As a user I want to browse and take up products so that I'm able to use my selected product

US4	View Products	As an existing user I want to view available products so that I can select a product to take up
US5	Check if eligible	As an existing user I want to have my eligibility checked so that I can take up a product
US6	Open Account	As an existing user I want to have my account opened so that I can take up a product
US7	Do customer checks	As an existing user I want to have checks done so that I can take up a product
US8	Generate Contract	As an existing user I want to have my contract generated so that I am aware of the product terms

5 Functional Requirements

User Story	Functional Requirement	System Requirements	
US1 – Log In	FR1 A user must be able to log into product shop	SR1	- A customer information store must hold customer profile information to facilitate the log in, and a customer should have a unique identifier - This must be exposed via a web service - Any user not logged in can trigger the login process, but only registered users must be able to log in successfully
US2 – Create Profile	FR2 An unregistered user must be able to create a profile	SR2	- New users must be able to be added to customer information store using a web service - Previously registered users with the same identifier should be prevented from re-registering
US3 – Take Up Product	<i>The requirements for this user story are broken down in US4 - US8 below</i>		
US4 – View Products	FR3 A user must be able to view a list of available products	SR3	- A product database must hold details for each product - The product list should be query-able via a web service
	FR4 A user must be able to select a product for further information	SR4	A product's details must be query-able via a web service
US5 – Check if Eligible	FR5 The system must verify the user's eligibility for their selected product	SR5	- Product eligibility criteria must be stored in the product database and access via a web service - Customer information must be queried and checked against the product eligibility criteria to determine eligibility - A customer would be considered eligible if the following criteria match between the customer and the product:

			○ The customer must have one of the qualifying accounts currently opened against their profile ○ The customer's Customer Type should match the customer type of the product that they are trying to take up.
US6 – Open Account	FR6 An account must be provisionally opened for the user	SR6	• Integration must be done with the product house system that owns the selected product to open the customer's account in a pending state
US7 – Do Customer Checks	FR7 The system must initiate one or more checks to determine if the customer can take up the product	SR7	• A request queue and processor must be implemented to store and process fulfilment request messages • Rules must be implemented to determine the fulfilment type (A, B, or C) based on the product, and indicated on the messages for the correct set of checks to be initiated • A response message processor must relay the results of the checks back to the initiating system
		SR8	• Integration must be done into a KYC service to do customer KYC verification This service is documented in authentication Service This service generates a JWT Token to be used for system-to-system integrations. You will generate the token with your applications username & password. Authentication Mechanism: Basic Auth URL: localhost:8080 Request: <pre>curl --location --request POST 'http://localhost:8080/token' \ --header 'Authorization: Basic dXNlcjpwYXNzd29yZA==' \ --header 'Cookie: XSRF-TOKEN=f41ef9ef-75f2-49e8-8142-97e6512da66d'</pre> Response:

			eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3VlJoidXNlciIsImV4cCI6MTY5MDQ1Mjk5MywiaWF0IjoxNjkwNDQ5MzkzfQ.zy2ATKNuU_sVlwqQyvYNCdCcexrVH8epjAtGddaCQWhkT9kOfkO4KR9Qoj0EjkMIQRsyQL90S5otjtfiRQT8_eLPCoWyb_ByP5nHzEFP5YwVRYzAmU8fTq-L6b1wUsFVF3HjzN1XVmNsF1zT5FH-pl8asTWdv83B3y-wDbtMd_FYEKjI3gnR_dsrtU0-L9UXf41gRdPMWXmNLadT6au0m7mSU3d37IEiIXdCNorFnmcN2mp6c34UKdJUXdlyltb0e9F81N-rcqPKrtjtoTnOEItgdgTZJ13Ca_q3x9iO_8xfFhfz4AvjJOQn1_qswr9JxcHyXXbJG6F9PDhg8MIIDJQKYC Service.
		SR9	- Integration must be done into the department of home affairs to do the following identity checks: - Marital status - Duplicate ID - Living/Deceased status This service is documented in Department of Home Affairs Service.
		SR10	Integration must be done into a fraud service to confirm that the customer has not participated in fraudulent activities This service is documented in Error! Reference source not found.
		SR11	Integration must be done into a Credit Check service to confirm the customer is in good credit standing This service is documented in Error! Reference source not found.
US8 – Generate Contract	FR8 A contract must be generated and returned to the user	SR12	- If customer checks all pass, a PDF contract must be generated from the customer's profile data and product information - If any customer check failed, a failure reason should be provided to the customer

6 Process Flow

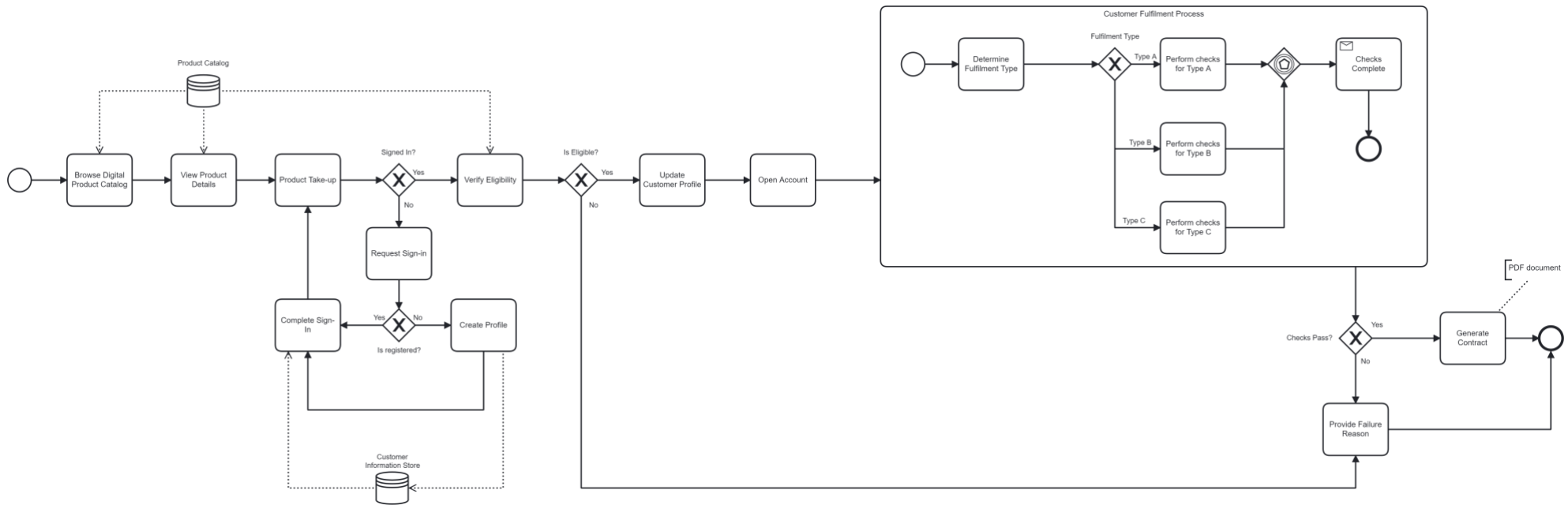


Figure 6-1BPMN Diagram of the Product Onboarding and Fulfilment Flow

6.1 FLOW DESCRIPTION

Basic flow scenario	1	The user selects and opens the product catalogue to browse available products, the system retrieves and displays available products from the Product Catalogue database.
	2	The user selects a product and the system displays details for the product, such as: • Description • Eligibility Criteria • Pricing • Benefits These details are retrieved from the Product Catalogue database.

	3	The user opts to take-up the selected product, which first checks if the user is signed in
	4	If the user is signed in their product eligibility is verified
	5	If the user is eligible, the customer profile is updated with their new product, in a pending state
	6	The account associated with the product is then opened, in a pending state
	7	The customer fulfilment process is then triggered
	8	Based on the product, the fulfilment type is queried. Currently there are two types: Type A and Type B and Type C. If multiple products are taken up that have differing flows, the flow with the most checks is the one that should be executed. These checks only need to be done once and not once pre product taken up.
	9	If Type A, the checks for Type A are done against the customer's profile: KYC check This is done asynchronously with a workflow process while the user is informed that feedback will be provided when complete.
	10	The system receives an indication that the checks are complete
	11	If the checks have passed, a product contract is generated for the user to view in PDF format
Variation: Not Signed In	4.1	The user is not signed in, present the sign-in screen
	4.2	The user is registered and enters their credentials
	4.3	The system verifies their credentials from the Customer Information Store and completes the sign in process
	4.4	The system redirects the user to Product Take-up, where the user confirms their choice
Variation: Not Registered	4.2.1	The user is not registered and is directed to create a profile. The Customer Information Store stores this new profile
Variation: Not Eligible	5.1	If the user is not eligible, the user is informed and the process ends
Variation: Type B	9.1	If Type B, the checks for Type B are done against the customer's profile: - KYC Check - Fraud Check - Living Status Check (DHA) - Duplicate ID Check
Variation: Type C	9.1.1	If Type C, the checks for Type C are done against the customer's profile: - KYC Check - Fraud Check - Living Status Check (DHA) - Duplicate ID Check (DHA)

		• Credit Check • Marital Status Check (DHA)
Variation: Checks Fail	11.1	If one of the checks fails, the user is informed and the process ends

7 Deployment

All the applications will be deployed as Containers on a platform to be determined in the future. For now, you will just need to containerise your application using a Docker file. The supporting services that have been built out have a Docker Compose file that can be used to start up the environment. You should add your services to this compose file and all the containers should be able to run using the single compose file.

8 Service Specifications

These services are not maintained by various other teams within the organisation. Reliability, as a result, can be unpredictable. We have seen that these services work approximately 80% of the time. You need to ensure that your code is robust enough to handle this scenario, and ensure that the customers application can still be processed. The OpenAPI specs and WSDL files for the services will be shared along with this requirements document.

8.1 AUTHENTICATION SERVICE

This service generates a JWT Token to be used for system-to-system integrations. You will generate the token with your applications username & password.

Authentication Mechanism: Basic Auth

URL: localhost:8080

Request:

```
curl --location --request POST 'http://localhost:8080/token' \  
--header 'Authorization: Basic dXNlcjpwYXNzd29yZA==' \  
--header 'Cookie: XSRF-TOKEN=f41ef9ef-75f2-49e8-8142-97e6512da66d'
```

Response:

eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmbWVudC0iMTY5MDQ1Mjk5MywiaWF0IjoxNjkwNDQ5MzkzfQ.zy2ATKNuU_sVlwqYyvYNCDcCexrVH8epjAtGddaCQWhkT9kOfko4KR9Qoj0EjkMIQRsyQL90S5otjtftRQT8_eLPCoWyb_ByP5nHzEFp5YwVRyZAmU8fTq-L6b1wUsFVF3HjzN1XVmNsF1zT5FH-pl8asTWdv83B3y-wDbtMd_FYEKjI3gnR_dsrtUO-L9UXf41gRdPMWXmNLADt6au0m7msU3d37IEilXdCNorFnmcN2mp6c34UKdjUXdlYltb0e9F81N-rcqPKrtjtoTnOEldtgTZJ13Ca_q3x9iO_8xfHfhz4AvJJQQn1_qswr9JxcHyXXBJG6F9PDhg8MIIDJQ

8.2 KYC SERVICE

Service Type: REST

Authentication Mechanism: Bearer Token

Base URL: localhost:8081

8.2.1 KYC Check

This service will determine if the requested customer is KYC compliant. It takes in a customer id and returns 3 indicators for the users KYC Status, namely:

- Primary Indicator
- Secondary Indicator
- Tax Compliance

In our scenario, a user is only able to take up our products if their primary indicator is a "true" – indicating that we do know our customer - and if the Tax Compliance is either AMBER or GREEN.

Request:

```
curl --location 'http://localhost:8081/kyc/123' \  
--header 'Authorization: Bearer  
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3ViljoidXNlcilImV4cCI6MTY5MDQ1MjUzNywiaWF0IjoxNjkwNDQ4O  
TM3fQ.vAln15u7cRiilKphFvjIvYb1K0JKen2yy7t4bvFVXKSahLoJ6833PV6ZjF8UW191dofrnicWkaomxn21-  
3hX932hgQMeG7KTq-oUUnn2QN-opGCTA492JdY-p4UwqWLv6VCTakUDmbCghSh9j-  
S1RbwZ8l1FzLP20ToTNNxd8dKnUZyFwXcntJl3G-o_q0H3sgElG6iZrP-  
tV4Y4tVZMQzEVfyAWieaB053lwbyymRL1gq3f07bAIU1Lf2M1Xn4--  
ChXwnwh4czzDRmlw9g_jgXKNLMNaO8w_7_1_05Im7dH8qHC8oD1XWLFw48AOqjHK9TirQDCpGY3D1ag'
```

Response:

```
{  
  "primaryIndicator": false,  
  "secondaryIndicator": false,  
  "taxCompliance": "green"  
}
```

8.3 DEPARTMENT OF HOME AFFAIRS SERVICE

This service proxies webservice calls to the department of home affairs. As a result, it takes in the id number of a customer and not the customer id in all of its requests. You will need get the id number for a specific customer from the Customer Information Store service.

Service Type: REST

Authentication Mechanism: Bearer Token

URL: localhost:8082

8.3.1 Marital Status Check

This service takes in an RSA ID number and returns the users current marital status, as well as a list of previous marital status' that user may have. If a customer selects a product that requires a Marital Status check, the system needs to confirm that the current status is "married".

Request:

```
curl --location 'http://localhost:8082/status/marital/4006183645875' \  
--header 'Authorization: Bearer  
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmliwic3ViljoidXNlcilImV4cCI6MTY5MDQ1MjUzNywiaWF0IjoxNjkwNDQ4O  
TM3fQ.vAln15u7cRiiIKphFvjiVYb1K0JKen2yy7tI4bvFVXKSahLoJ6833PV6ZjF8UW191dofrnicWkaomxn21 -  
3hX932hgQMeG7KTq-oUUnn2QN-opGCTA492JdY-p4UwqWLv6VCTakUDmbCghSh9j-  
S1RbwZ8l1FzIP20ToTNNxd8dKnUZyFwXcntjI3G-o_q0H3sgElG6iZrP-  
tV4Y4tVZMQzEvfyAWieaB053lwbymRL1gq3f07bAIU1Lf2M1Xn4--  
ChXwnwh4czrDRmlw9g_jgXKNLMNaO8w_7_1_05Im7dH8qHC8oD1XWLfW48AOqjHK9TirQDCpGY3D1ag'
```

Response:

```
{
  "currentStatus": {
    "status": "Single",
    "effectiveFrom": "2023-07-30",
    "effectiveTo": ""
  },
  "previousStatus": null
}
```

8.3.2 Duplicate ID Status Check

This service takes in an ID number and returns whether or not the user has a duplicate Identity Document issued. Due to the risks of having 2 Identity Documents, no customer that needs to do a Duplicate ID Check can have 2 ID's and you should validate that the response is false.

Request:

```
curl --location 'http://localhost:8082/status/duplicateId/4006183645875' \
--header 'Authorization: Bearer eyJhbGciOiJSUzI1NiJ9.eYlpc3MiOiJZZWxmlwiw3ViljoidXNlcilsImV4cCI6MTY5MDQ1MjUzNywiaWF0IjojNjkxNDQ0O
TM3fQ.vAln15u7cRiiIKphFvjiVYb1K0JKen2yy7tL4bvFVXKSahLoJ6833PV6ZjF8UW191dofrnicWkaomxn21-
3hX932hgQMeG7KTq-oUUnn2QN-opGCTA492JdY-p4UwqWLv6VCTakUDmbCghSh9j-
S1RbwZ8l1FzlP20ToTNNxd8dKnUZyFwXcntjI3G-o_q0H3sgElG6iZrP-
tv4Y4tVZMQzEVfyAWieaB053lwbyymRL1gq3f07bAIU1Lf2M1Xn4--
ChXwnwh4czrDRmIw9g_jgXKNLMNaO8w_7_1_05lm7dH8qHC8oD1XWLfW48AOqjHK9TirQDCpGY3D1ag' \
--header 'Cookie: XSRF-TOKEN=f41ef9ef-75f2-49e8-8142-97e6512da66d'
```

Response:


```
{
  "hasDuplicateId": false,
  "duplicateIdIssueDate": ""
}
```

8.3.3 Living / Deceased Status Check

This service takes in an ID number and returns if a person is alive or not. For obvious reasons, you should not allow a person that is deceased to take up a new product.

Request:

```
curl --location 'http://localhost:8082/status/living/4006183645875' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiIiwic3ViIjoiaXNlcilzImV4cCI6MTY5MDQ1MjUzNywiaWF0IjoxNjkwNDQ4O
TM3fQ.vAln15u7cRiilKphFvjIvYb1K0JKen2yy7tI4bvFVXKSahLoJ6833PV6ZjF8UW191dofrnicWkaomxn21-
3hX932hgQMeG7KTq-oUUnn2QN-opGCTA492JdY-p4UwqWLv6VCTakUDmbCghSh9j-
S1RbwZ8l1FzIP20ToTNNxd8dKnUZYFwXcntjI3G-o_q0H3sgElG6iZrP-
tV4Y4tVZMQzEVfyAWieaB053lwbyymRL1gq3f07bAIU1Lf2M1Xn4--
ChXwnwh4cZrDRmlw9g_jgXKNLMNaO8w_7_1_05lm7dH8qHC8oD1XWLFw48AOqjHK9TirQDCpGY3D1ag' \
--header 'Cookie: XSRF-TOKEN=f41ef9ef-75f2-49e8-8142-97e6512da66d'
```

Response:

```
{
  "livingStatus": "Alive",
  "deceasedDate": ""
}
```

8.4 CREDIT CHECK SERVICE

This service calculates a customers' credit status. We don't want to provide people who cannot manage their credit with services as this would increase our risk profile above an acceptable level.

Service Type: SOAP

Authentication Mechanism: None

URL: localhost:8083

8.4.1 Credit Check

This service takes in a customerId and returns the credit check status for a customer. The service will return the credit status as either RED, AMBER or GREEN. In order to open a new product, a customer cannot be in a RED credit status.

Request:

```
curl --location 'http://localhost:8083/CreditCheck' \  
--header 'Content-Type: application/xml' \  
--header 'Authorization: Basic dXNlcjpwYXNzd29yZA==' \  
--data '<soapenv:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" \  
xmlns:xsd="http://www.w3.org/2001/XMLSchema" \  
xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" \  
xmlns:ent="entelect:devcamp:creditcheckservice">  
  <soapenv:Header/>  
  <soapenv:Body>  
    <ent:CreditCheck soapenv:encodingStyle="http://schemas.xmlsoap.org/soap/encoding/">  
      <customerId xsi:type="xsd:int">123</customerId>  
    </ent:CreditCheck>  
  </soapenv:Body>  
</soapenv:Envelope>'
```

Response:

```
<s:Envelope xmlns:s="http://schemas.xmlsoap.org/soap/envelope/" \  
xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">  
  <s:Body>  
    <CreditCheckResponse xmlns="entelect:devcamp:creditcheckservice">  
      <CreditCheckResult>RED</CreditCheckResult>  
    </CreditCheckResponse>  
  </s:Body>  
</s:Envelope>
```

8.5 FRAUD SERVICE

This service is currently being built by the fraud team. The organisation has agreed that they will be developing using contract-first development. You can mock this service out in your environment or application based on the WSDL that has been provided.

Service Type: SOAP

Authentication Mechanism: Basic Auth

URL: NA

8.5.1 Fraud Check

This service is used to check if a customer has been suspected or convicted of committing fraud. We do not want to accept the risk of selling any of the included products to a customer that has been or is suspected of committing fraud.

Request:**Response:**

8.6 CUSTOMER INFORMATION STORE

The Customer Information Store manages the information for each customer. This includes basic personal information, products linked to the customer, contracts that the customer has signed, etc.

Service Type: REST

Authentication Mechanism: Bearer Auth

URL: localhost:8084

8.6.1 Fetch Customers

This operation will fetch all the customers currently existing within the Customer Information Store service.

Request:

```
curl --location 'http://localhost:8084/v1/customers' \  
--header 'Authorization: Bearer eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwlic3ViljoidXNlcilImV4cCI6MTY5MDg0MjMwNCwiaWF0IjoxNjkwODM4NzA0fQ.UKTNCsYFWfw8hba2MkiRUIU4__ZAmKqzD1BhJP65adWk4UQJeA1C281vVdTZ6sDrad_OaYh7dTuRtBd_ay0WCbKUcm7lGrMm93bKdLJMEOpF9Kuqj3k8rZqmWeH05kZ6aB_39AgicDNyLx-E6EGjhLNrlsbMO-QcdxuPVFEmvCSFDQqatbPB5Wh0mkGF2aCDxD0yM7GT0q-QOj7A1VKtjedKMQo5ekYMzMjFryYAMqqjVA77a7jZUwasCmm6-dZr2hte0oSy2Lg7HuV6iPRRELcKspzBuk5V1FNCXYi0b_2VsXkGEr_HAq1FTMUvWEYdbW00YQAFAlmGYbqFaxwxcw'
```

Response:

```
[  
  {  
    "id": 1,  
    "username": "user",  
    "password": null,  
    "firstName": "Jesse",  
    "lastName": "Leresche",  
    "customerTypeId": null,  
    "customerType": {  
      "id": 2,  
      "name": "Sample Type",
```

```
    "description": "Sample Type Description"
  },
  "customerAccounts": []
}
]
```

8.6.2 Fetch Customer Information

This service will return the basic customer information for a given customer id. This will be details like First Name, Last name, email, customer type etc.

Request:

```
curl --location 'http://localhost:8084/v1/customer/1' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3ViIjoidXNlcilImV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5MjA1fQ.gtaxaQd8ANyN-
slaBFq5QfhwOpynwT09q0GtVcZmIj9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1JyI22ZEsXNmreBlxTiw
mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-
i21IXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-
ckckuSojWzE4bWAbKtgdowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWyuppLDGZF8A5MCoVx5ZrGeAn
kNm7006OoMipmo-nzZT2ulQQVw'
```

Response:

```
{
  "id": 1,
  "username": "user",
  "password": null,
  "firstName": "Jesse",
  "lastName": "Leresche",
  "idNumber": "0001175432082",
  "customerTypeId": null,
  "customerType": {
    "id": 2,
    "name": "Sample Type",
    "description": "Sample Type Description"
  },
  "customerAccounts": []
}
```

8.6.3 Register Customer

This service will allow you to register a new customer with the customer information store.

Request:

```
curl --location 'http://localhost:8084/v1/customer/' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3ViljoidXNlcilslmV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5MjA1fQ.gtaxaQd8ANyN-
slaBFq5QfhwOpynwT09q0GtVcZmlJ9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1Jyl22ZEsXNmreBlxTiw
mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-
i21lXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-
ckckuSojWzE4bWAbKtgdDowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWYupplDGZF8A5MCoVx5ZrGeAn
kNm7006OoMipmo-nzZT2ulQQVw' \
--data-raw '{
  "username": "Sas.vanderwest@entelect.co.za",
  "password": "SasLovesSpring",
  "firstName": "Sas",
  "lastName": "Van Der Westhuizen",
  "customerTypeId": 3,
  "idNumber": "9205157654082"
}'
```

Response:

```
{
  "id": 8,
  "username": "Sas.vanderwest@entelect.co.za",
  "password": null,
  "firstName": "Sas",
  "lastName": "Van Der Westhuizen",
  "idNumber": "9205157654082",
  "customerTypeId": null,
  "customerType": null,
  "customerAccounts": null
}
```

8.6.4 Fetch Account Types

This service will return the account types that have been configured in the system.

Request:

```
curl --location 'http://localhost:8084/v1/accountTypes' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3ViljoidXNlcilslmV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5MjA1fQ.gtaxaQd8ANyN-
```

```
slaBFq5QfhwOpynwT09q0GtVcZmlJ9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1Jyl22ZEsXNmreBlxTiw  
mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-  
i21lXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-  
ckckuSojWzE4bWAbKtgddowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWyupplDGZF8A5MCoVx5ZrGeAn  
kNm7006OoMipmo-nzZT2ulQQVw'
```

Response:

```
[  
  {  
    "id": 3,  
    "name": "checking account",  
    "description": "foo"  
  }  
]
```

8.6.5 Add Account Types

Request:

```
curl --location 'http://localhost:8084/v1/accountTypes' \  
--header 'Content-Type: application/json' \  
--header 'Authorization: Bearer  
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3ViljoidXNlcilImV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5M  
jA1fQ.gtaxaQd8ANyN-  
slaBFq5QfhwOpynwT09q0GtVcZmlJ9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1Jyl22ZEsXNmreBlxTiw  
mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-  
i21lXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-  
ckckuSojWzE4bWAbKtgddowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWyupplDGZF8A5MCoVx5ZrGeAn  
kNm7006OoMipmo-nzZT2ulQQVw' \  
--data '{  
  "description": "foo",  
  "name": "checking account"  
}'
```

Response:

```
{  
  "id": 3,  
  "name": "checking account",  
  "description": "foo"  
}
```

8.6.6 Add Account Types to Customer

This service allows you to link a configured account type to the given customer via the customer id. It has no response body and just returns a 204 status code.

Request:

```
curl --location 'http://localhost:8084/v1/customer/1/accounts/3' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmliwic3ViljoidXNlcilslmV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5MjA1fQ.gtaxaQd8ANyN-slaBFq5QfhwOpynwT09q0GtVcZmlJ9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1Jyl22ZEsXNmreBlxTiw mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-i21IXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-ckckuSojWzE4bWAbKtgddowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWyupplDGZF8A5MCoVx5ZrGeAn kNm7006OoMipmo-nzZT2ulQQVw' \
--data '{
  "description": "Foo",
  "name": "Customer Checking Account"
}'
```

Response:

No Content

8.6.7 Fetch Customer Documents

This service will fetch the list of documents for a given customer id.

Request:

```
curl --location 'http://localhost:8084/v1/customer/1/documents' \
--header 'Authorization: Bearer eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmliwic3ViljoidXNlcilslmV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5MjA1fQ.gtaxaQd8ANyN-slaBFq5QfhwOpynwT09q0GtVcZmlJ9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1Jyl22ZEsXNmreBlxTiw mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-i21IXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-ckckuSojWzE4bWAbKtgddowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWyupplDGZF8A5MCoVx5ZrGeAn kNm7006OoMipmo-nzZT2ulQQVw'
```

Response:


```
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3ViljoidXNlcilImV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5MjA1fQ.gtaxaQd8ANyN-
slaBFq5QfhwOpynwT09q0GtVcZmlJ9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1Jyl22ZEsXNmreBlxTiw
mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-
i21lXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-
ckckuSojWzE4bWAbKtgddowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWyupplDGZF8A5MCoVx5ZrGeAn
kNm7006OoMipmo-nzZT2ulQQVw' \
--data '{
  "name": "Test Name",
  "description": "description"
}'
```

Response:

```
[
  {
    "id": 2,
    "name": "Sample Type",
    "description": "Sample Type Description"
  }
]
```

8.6.10 Create Customer Types

This service allows you to create a new customer type within the system.

Request:

```
curl --location 'http://localhost:8084/v1/customerTypes' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3ViljoidXNlcilImV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5MjA1fQ.gtaxaQd8ANyN-
slaBFq5QfhwOpynwT09q0GtVcZmlJ9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1Jyl22ZEsXNmreBlxTiw
mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-
i21lXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-
ckckuSojWzE4bWAbKtgddowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWyupplDGZF8A5MCoVx5ZrGeAn
kNm7006OoMipmo-nzZT2ulQQVw' \
--data '{
  "name": "Test Name",
  "description": "description"
}'
```

Response:

```
{
  "id": 9,
  "name": "Test Name",
  "description": "description"
}
```

8.6.11 Update Customer with Customer Type

A customer can only be 1 customer type at a time – i.e. they cannot be a corporate customer as well as an individual customer simultaneously. As a result, you are able to update the Customer Type that a Customer is associated with by calling this service.

Request:

```
curl --location --request PUT 'http://localhost:8084/v1/customer/1/customerTypes/9' \
--header 'Content-Type: application/json' \
--header 'Authorization: Bearer
eyJhbGciOiJSUzI1NiJ9.eyJpc3MiOiJzZWxmiwic3ViljoidXNlcilImV4cCI6MTY5MDg0MjgwNSwiaWF0IjoxNjkwODM5MjA1fQ.gtaxaQd8ANyN-slaBFq5QfhwOpynwT09q0GtVcZmlJ9wWvNwCBeMo2aq8bWnn4OFf9D3V57NuQyVT67yS1Jyl22ZEsXNmreBlxTiw
mHWWVg8l6L8_jfa3aT4p2VTkimgnhnzeP72LRaXbpPwaHd9SE-i21lXjAWfnq6di3dFlupCPeB1BkELy3k66UoRE4GQq_oMjKnwulRsVoFwx2gYe3g2Jqkp1RiRjy0R1-
ckckuSojWzE4bWAbKtgdDowCG3PkutlpTHLWofFn_RHw1OQ7AUKww0iYwZ4oLWYupplDGZF8A5MCoVx5ZrGeAn
kNm7006OoMipmo-nzZT2ulQQVw' \
--data '{
  "document": "string"
}'
```

Response:

204 NO CONTENT

8.7 SERVICE BUS

Due to the various checks that need to happen in the fulfilment process, and the possibility for error, the fulfilment process should be completed asynchronously. The approved technologies in the bank are to use a JMS store to handle asynchronous communication. In this case the technology used should be RabbitMQ. Given the fact that these messages will only be consumed by your 2 services, you will need to define the structure of these yourself.

9 Databases

9.1.1 ERD's

9.1.1.1 Product Catalogue

Below is the ERD for the Product Catalogue. This is the design that the DBA Team has constructed for your application(s). Should you need to make any changes to the below you will need to consult with the DBA Team for approval. You should version your services appropriately to cater for changes that they may require further down the line in the SDLC.

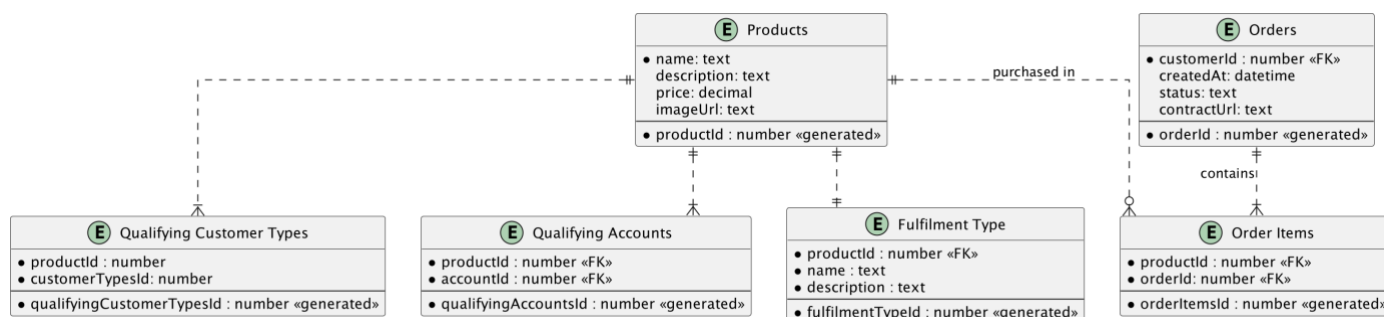


Figure 9-1 Product Catalogue ERD

9.1.1.2 Customer Information Store

The Customer Information Store has already been built out – the below ERD is simply to illustrate how all the data fits together.

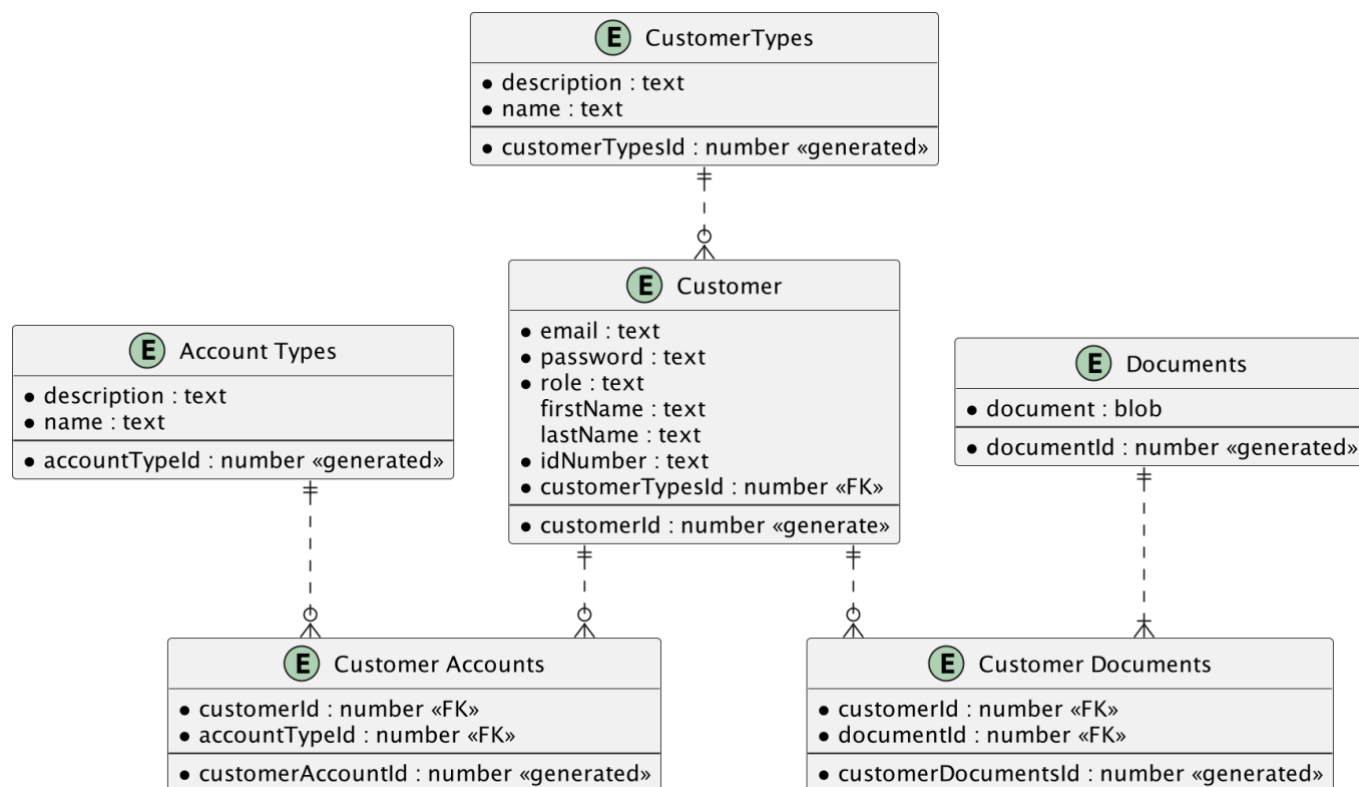


Figure 9-2 Customer Information Store ERD

10 Technical Setup

The starter project that will be shared with you will have the details on how to get the dependencies running on your machine. As mentioned previously, it is up to you to add in the 2 services that you have built into this docker-compose file.

NOTE: Any Engineer within the organisation should be able to use this docker-compose file to run the various applications. As such, your application needs to be built and pushed into the Entelect Harbor Docker Repository for anyone else to be able to access.

11 Reference Documents

Name	Document	Link/Document
Vitesh Jinabhai	Use Case Diagram (diagrams.net)	 v1 Banking Use Case.svg
Vitesh Jinabhai	Process Flow (Camunda)	 Product Take-up Process.bpmn

12 Document Details

12.1 DISTRIBUTION LIST

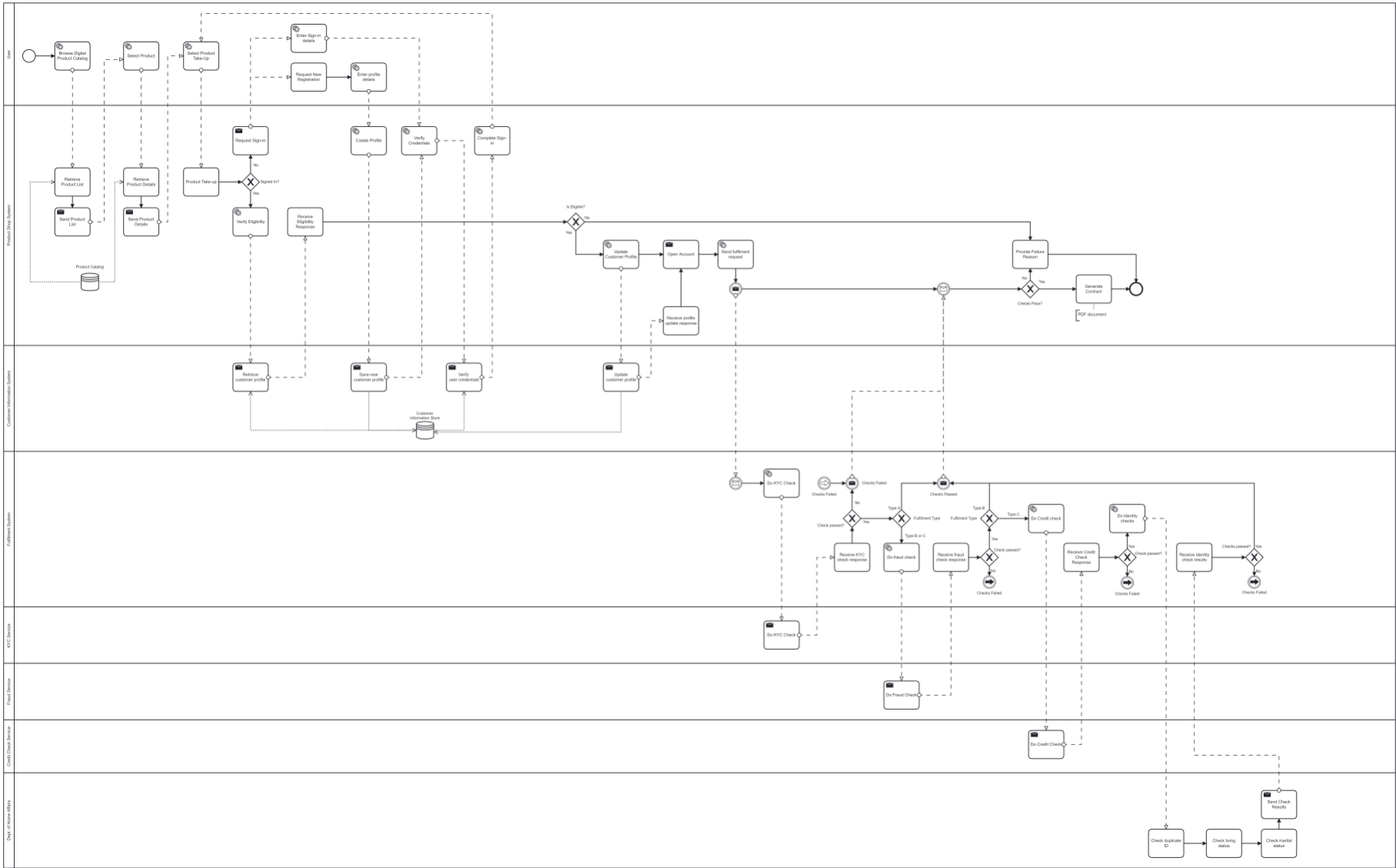
Name	Role	Sign-off Review Info
	Business Owner	Sign-off
	Business Lead	Sign-off
	Team Technical Lead	Sign-off

12.2 DOCUMENT HISTORY

Who	Date	Version	Description
Vitesh Jinabhai	2023-07-12	0.1	Document creation
Jesse Leresche	2023-07-27	1	Added in Service Details & Product Catalogue ERD
Jesse Leresche	2023-07-31	2	Added in Customer Information Store ERD & Service requests, Updated Product ERD
Jesse Leresche	2023-08-04	2.1	Added in Appendix B with Sample Product Data, updated fulfilment checks based on type.

Appendix A: Detailed Process Flow

A detailed process flow is provided below:



13 Appendix B: Product Sample Data

Sample Data for the types of products you can expect your system to cater for.

13.1 PRODUCT

Name	Description	Price	Fulfilment Type
Retail Short Term Insurance	Provides cover for short-term products for individuals - Electronics, Household Items, Jewellery, Cars etc.	R500	C
Retail Long-Term Insurance	Provides cover for longer term products individuals - household insurance, life insurance etc.	R1000	C
Commercial Short Term Insurance	Provides cover for short-term products for commercial entities - Printers, Company Cars, Theft, etc.	R5000	C
Commercial Long-Term Insurance	Provides cover for longer term products - office insurance, employee benefit insurance, etc.	R10000	C
Device Contract	Allows the customer to take out a device on contract - such as a phone, laptop etc.	R850	A
Short-Term Investment Product	Provides a way for customers to invest their money over a short period of time - 32 day fixed deposit etc.	R2500	B
Long-Term investment Product	Provides a way for users to invest their money over the long term - Retirement / Annuity Funds, Unit Trusts etc	R5000	B
Islamic Investment Product	Provides a way for Islamic customers to invest their money	R5000	B
VIP Investment Product	Provides an Investment product for VIP customers Over 150 Million Net-Asset Value	R20000	B

13.2 FULFILMENT TYPE

Name	Description
A	Smallest unit of checks. This just requires KYC to be completed.
B	Requires all of the checks in Type A, as well as the Fraud Check, Living Status Check and Duplicate ID Check
C	Requires all of the checks in Type B, as well as the Marital Status Check & Credit Check

13.3 QUALIFYING ACCOUNTS

Product Name	Account Types
Retail Short Term Insurance	Gold Cheque Account, Platinum Cheque Account, Signet Cheque Account, Islamic Cheque Account

Retail Long-Term Insurance	Gold Cheque Account, Platinum Cheque Account, Signet Cheque Account, Islamic Cheque Account
Commercial Short Term Insurance	SME Checking Account, Medium Enterprise Checking Account, Large Enterprise Checking Account
Commercial Long-Term Insurance	SME Checking Account, Medium Enterprise Checking Account, Large Enterprise Checking Account
Device Contract	Gold Cheque Account, Platinum Cheque Account, Signet Cheque Account, Islamic Cheque Account, Savings Account
Short-Term Investment Product	Gold Cheque Account, Platinum Cheque Account, Islamic Cheque Account
Long-Term investment Product	Gold Cheque Account, Platinum Cheque Account, Islamic Cheque Account
Islamic Investment Product	Islamic Cheque Account
VIP Investment Product	Signet Cheque Account

13.4 QUALIFYING CUSTOMER TYPES

Product Name	Customer Types
Retail Short Term Insurance	INDIVIDUAL
Retail Long-Term Insurance	INDIVIDUAL
Commercial Short Term Insurance	SOLE PROP, NON-PROFIT, CIPC
Commercial Long-Term Insurance	SOLE PROP, NON-PROFIT, CIPC
Device Contract	INDIVIDUAL, SOLE PROP, NON-PROFIT, CIPC
Short-Term Investment Product	INDIVIDUAL, SOLE PROP, NON-PROFIT, CIPC
Long-Term investment Product	INDIVIDUAL, SOLE PROP, NON-PROFIT, CIPC
Islamic Investment Product	INDIVIDUAL, NON-PROFIT
VIP Investment Product	INDIVIDUAL